



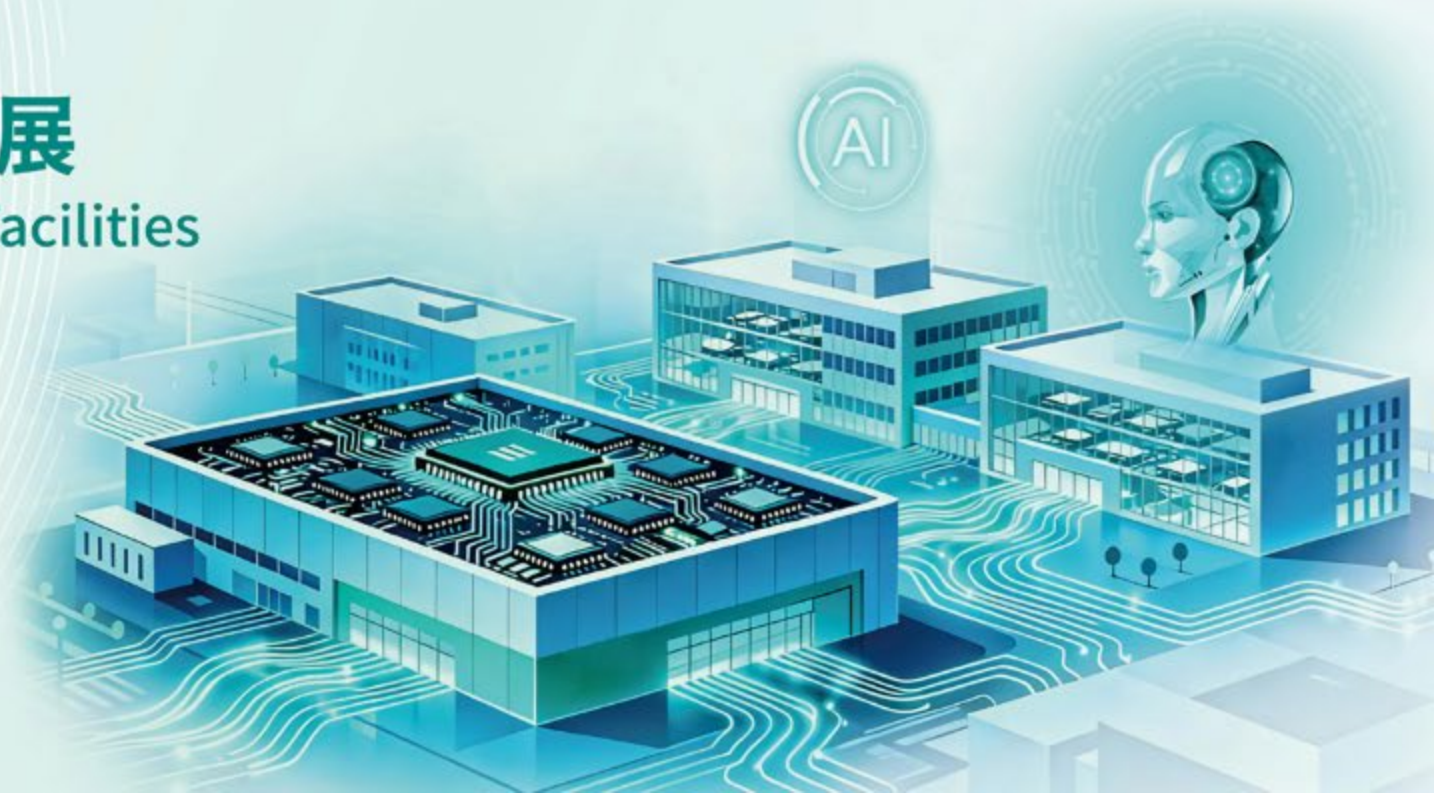
2026 高科技廠房設施國際論壇

High-Tech Facility International Forum 2026

以AI驅動創新加速 高科技廠房設施的未來發展

Accelerating the Future of High-Tech Facilities
Through AI-Driven Innovation

Thursday, September 3, 2026





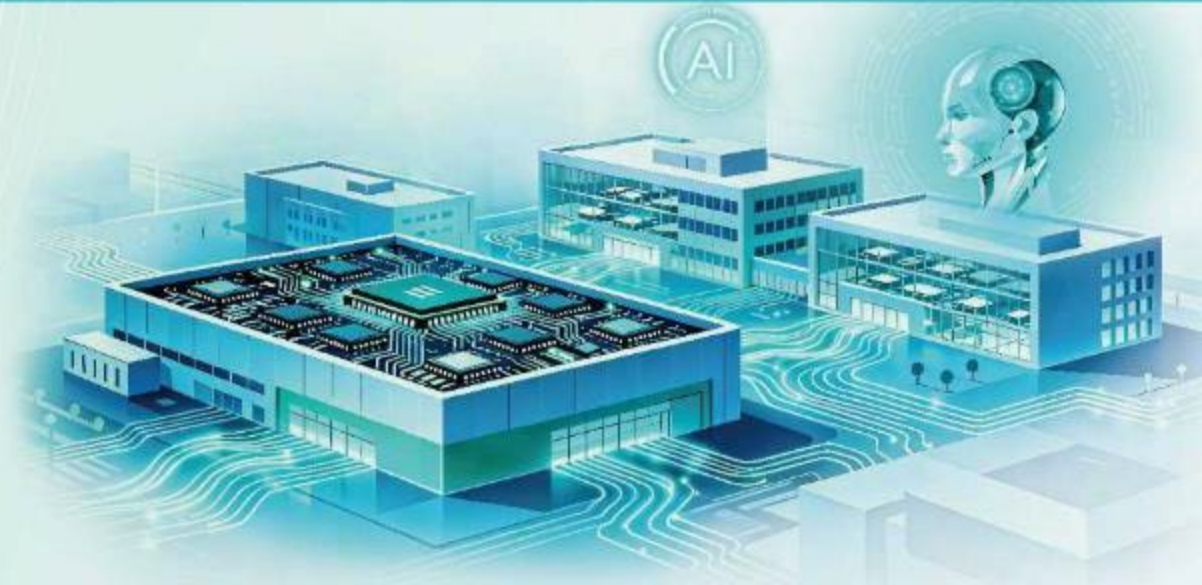
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Eyes in the Air

How AI-Driven UAV Redefine Indoor Inspection and 3D Progress Modeling in High-Tech Facilities



Sam Shih-Cheng Chien, PhD
Yankey Engineering Co., Ltd.



Industry Background & Construction Challenge

Stagnant Productivity Growth in Construction

- **Growth comparison:**

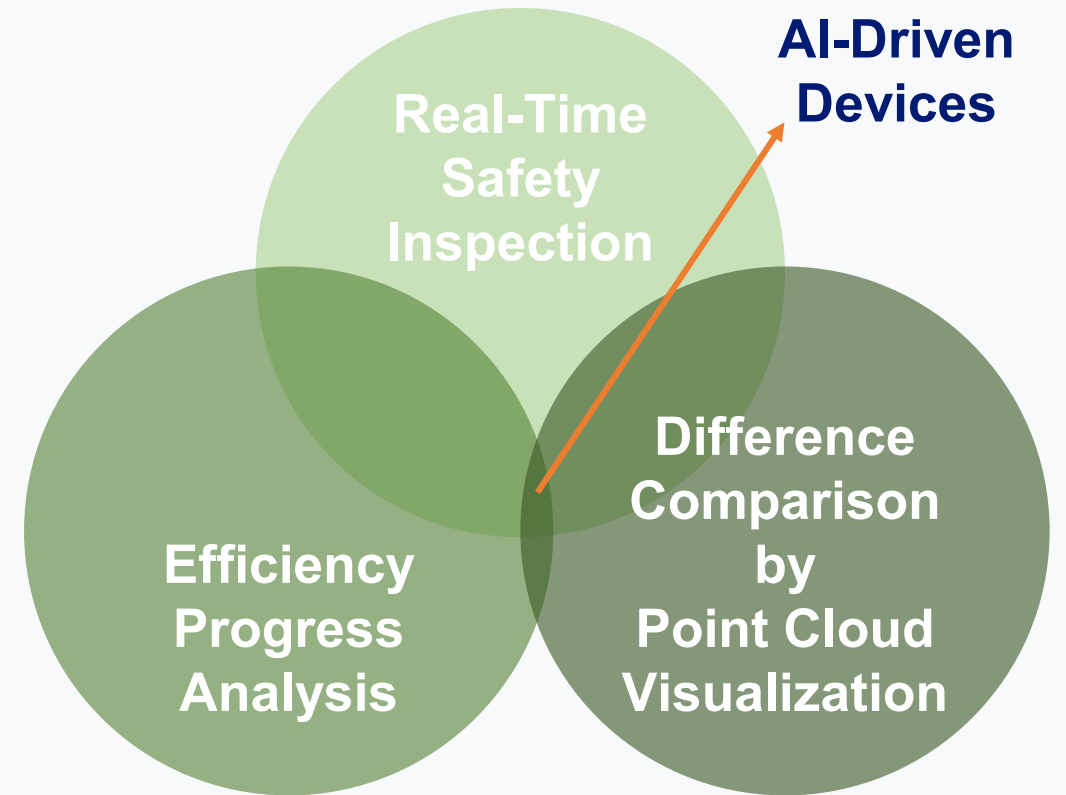
Construction 1% vs. Manufacturing 3.6%.^[1]
in 20-Year Productivity Growth



- **Local Structural Issues**
- **Inspection Efficiency Bottleneck**
- **Low Degree of Standardization**
- **Underinvestment in digital and automated Technology**

^[1] McKinsey Global Institute. (2017). Reinventing Construction: A Route to Higher Productivity. McKinsey & Company

Structural & Safety Challenges



YKE Point Cloud Technology Applications

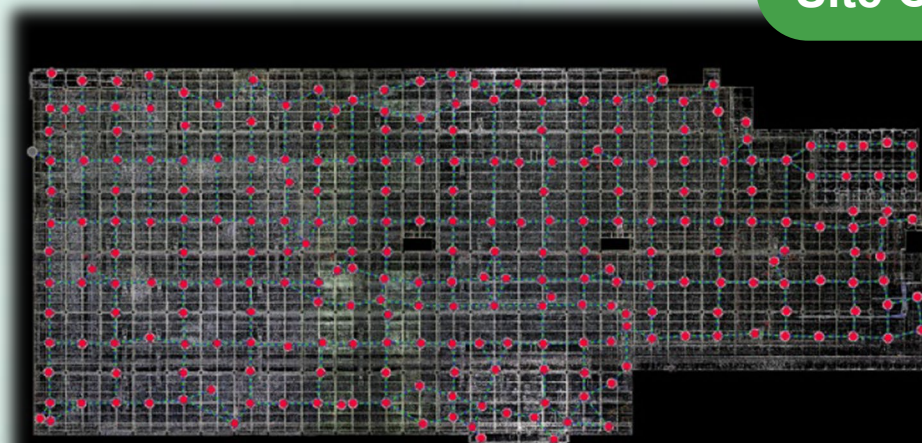
Colored Point Cloud



Point Cloud and BIM Model Overlay Comparison



Site Configuration



Point Cloud Registration Quality Check

Overall Quality

Error Results for Bundle 1

Setup Count:	266
Link Count:	371
Strength:	89 %
Overlap:	77 %
Global Bundle Error:	0.0006 m

Bundle Error
0.0026 m ✓

Overlap
77 % ✓

Strength
89 % ✓

Cloud-to-Cloud
0.0026 m ✓

Target Error
--

Max error of 0.0150 m.

Max error of 0.0200 m.

Error greater than 0.0200 m.

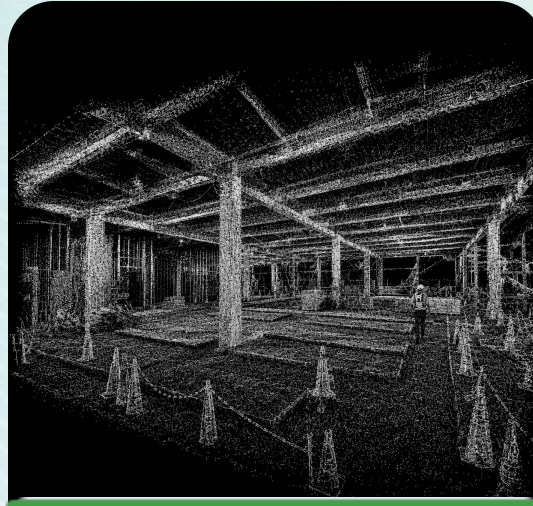
AI-Driven UAV Applications On-Sites

Real-time As-Built validation



Safety Behavior Recognition

AI vision to detect real-time behaviors



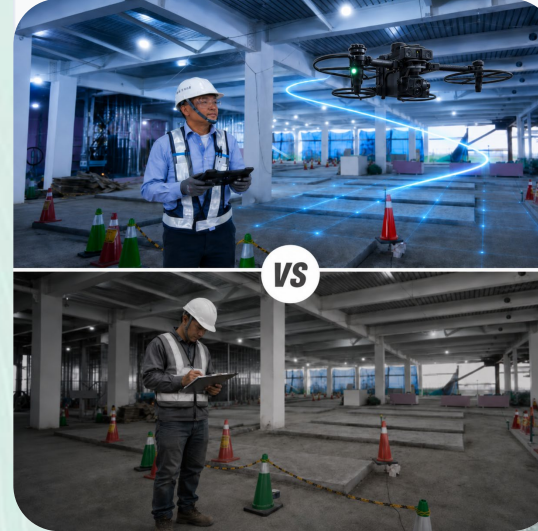
3D Modeling

Optical cameras rapidly generate 3D point cloud models integrated with BIM



Smart Inspection Platform

Regular automated inspections to cover blind spots



Efficiency Enhancement

Simultaneously for recognition & recording



AI-Driven UAV Core Modeling Technologies

Photogrammetry

01

- 3D Reconstruction from 2D Photos
- Centimeter-Level Accuracy

Point Cloud Visualization

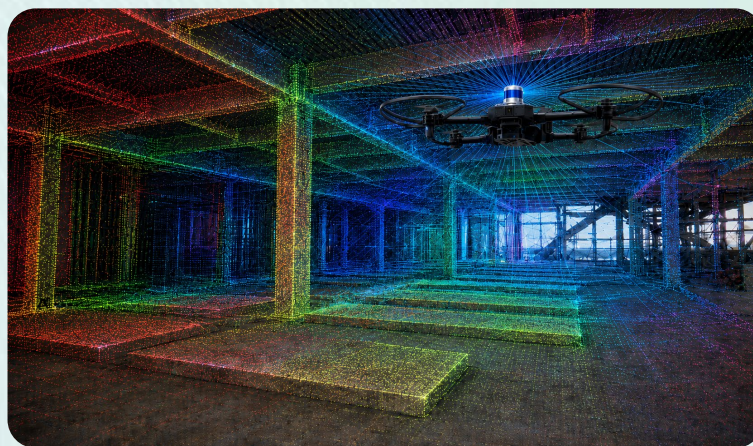
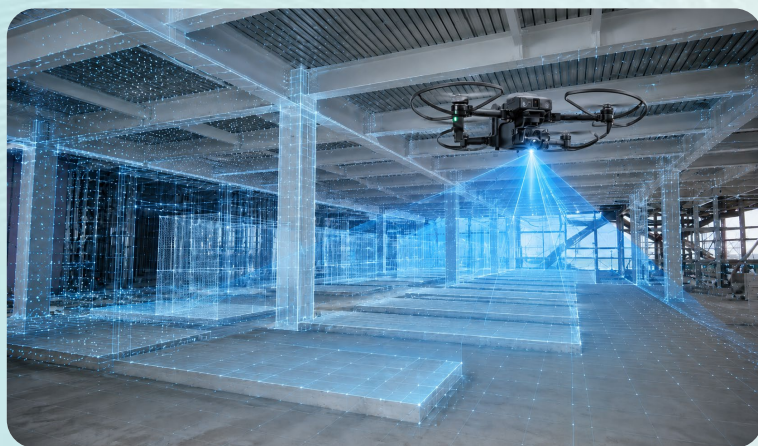
02

- High-Precision 3D Point Clouds
- Real-Time Earthwork Calculation

Integrated BIM Workflow

03

- UAV Data Collection →
- .las Point Cloud Export →
- Export to ReCap (.rcp) →
- Revit BIM Comparison ✓



Current Tech Bottlenecks & Practical Limitations

Bottleneck	Practical Limitation	Impact Assessment
Outdoor Modeling Only	Indoor flight/modeling is infeasible due to GPS denial	Critical Gap
Limited PPE Detection Accuracy	Suboptimal camera angles reduce detection of small items <i>e.g.</i> , hook	Technical Limitation
Intermittent Inspection	Patrols are periodic, lacking real-time, continuous alerting	Timeliness Limitation
Battery Life Limitation	15 min flight time restricts large-scale coverage	Hardware Limitation
Airspace Regulations	Complex application processes delay quick deployment	Compliance Limitation



Avilon as the YKE R&D Partner



Proprietary Technologies & Proven Track Record

- **Pioneering Made-in-Taiwan R&D**

In-house core technology development, tailored specifically for complex industrial and construction environments

- **Extensive Industry Track Record**

- Japanese automaker's smart factory warehouse in Thailand
- Partnered with MSI to develop a Mobile Autonomous Drone Charging Robot
- Deeply integrated with Taiwan Secom Co., Ltd.

- **High Customizability**



Indoor AI-Driven UAV Technical Highlights

Specification	Features & Description	Technical Advantage
Indoor Positioning	vSLAM Navigation: GPS-free, low-level infra deployment	Zero-Infrastructure Deployment
AI Edge Image Recognition	Smart EHS: Onboard customizable PPE & safety hazard detection	Smart Real-Time
Multi-Sensor Integration	High Expandability: Supports QR CODE, RFID, and thermal sensors	Multi-Task Inspection
Flight Endurance	Indoor Endurance: 30 - 40 min/session of stable flight	High Coverage
Edge Computing Power	High-Performance Processing: Onboard 15 TOPS chip for low-latency AI inference	Edge Intelligence



AI-Driven UAV Deployment Mission Scope

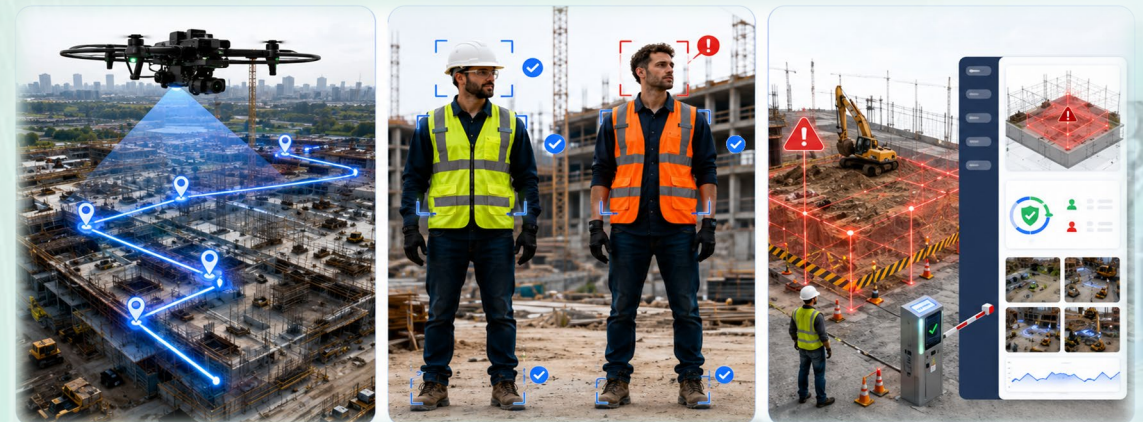
Construction Progress & Quality Tracking

- Rapid Full-Coverage Image Scanning
- BIM Model Verification & Alignment
- Digital Archiving & Site Documentation

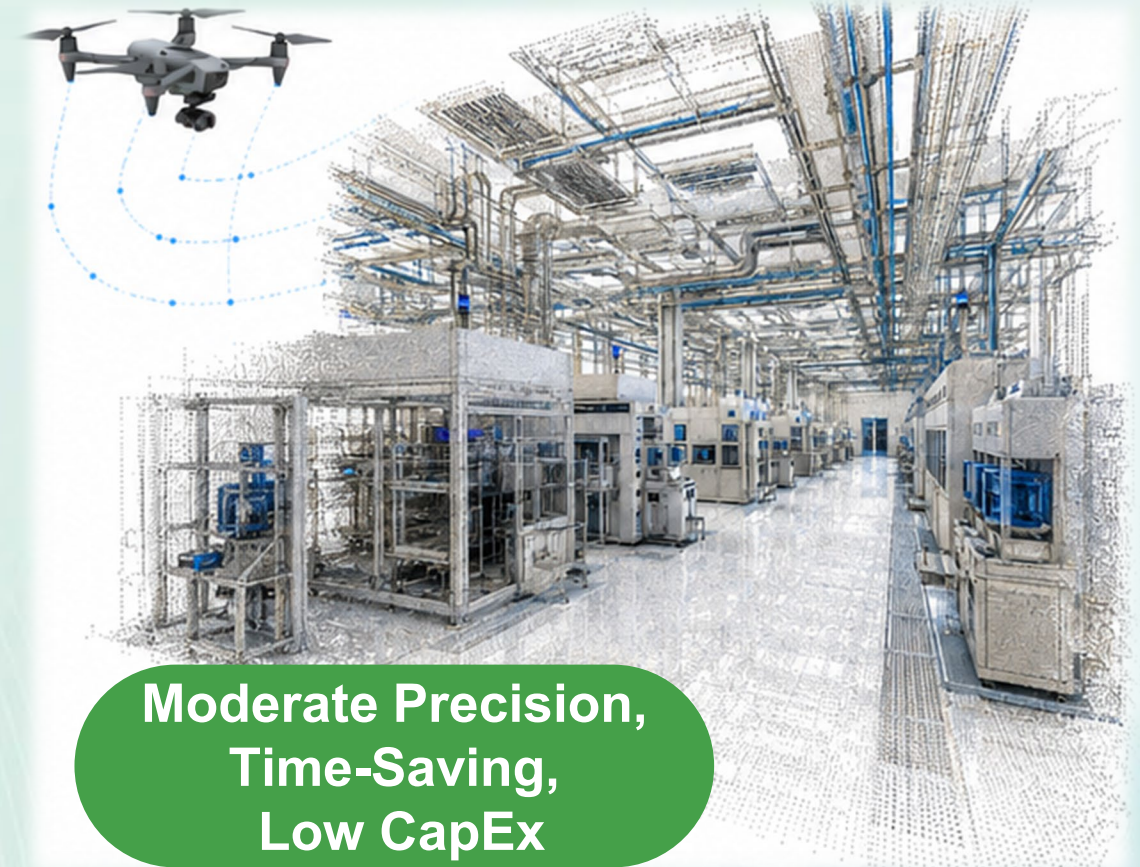
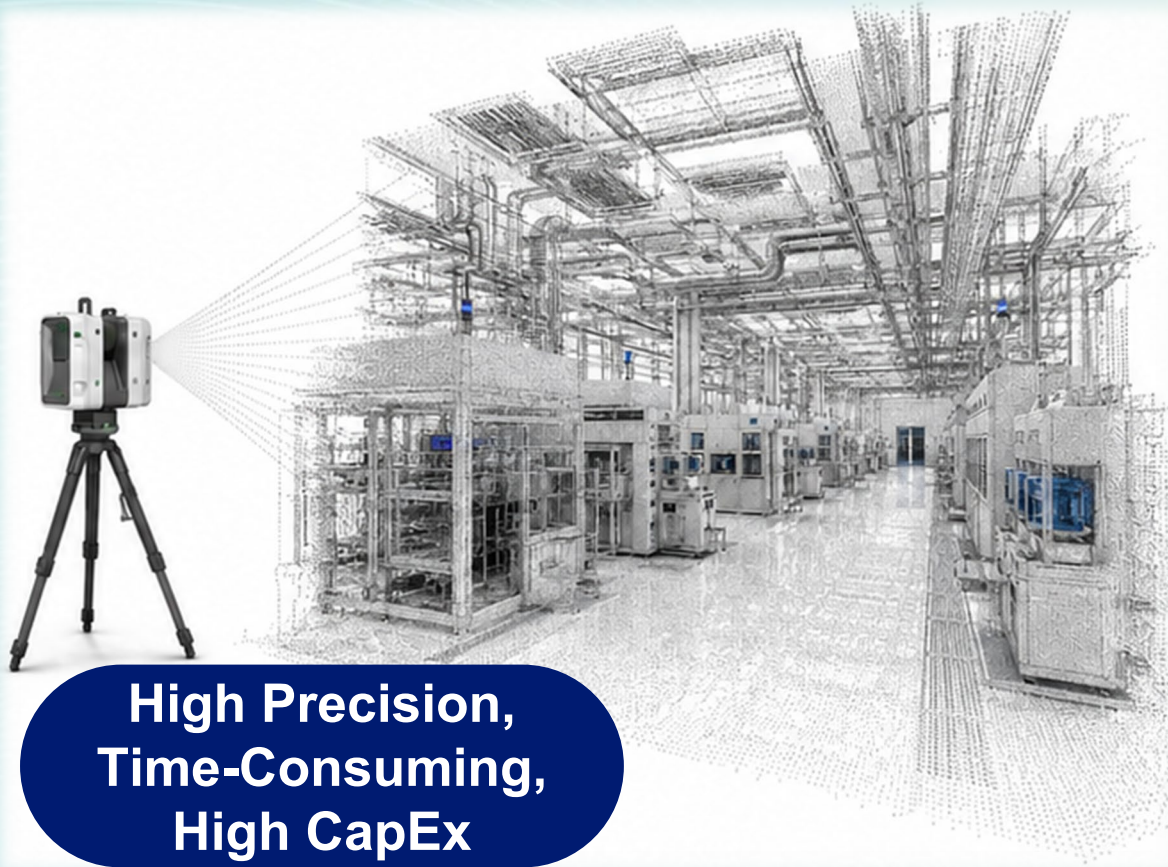


Smart Safety Patrol

- High-Precision Automated Inspection Routes
- PPE & Equipment Compliance Detection
- Hazardous Zone Access Control



TLS, Terrestrial Laser Scanner vs. UAV Photogrammetry



Yankey Engineering's Smart Inspection Initiative

Synergy & Execution

- Strategic Partnership
- Fixed & Mobile Synergies
- UAV Advantages
- Fixed Surveillance Benefits



Time-Based Event Logging U



Point Cloud Visualization

Forward Drone Viewer v2

YKE 洋基工程股份有限公司
YANKEY Engineering Co., Ltd.

FORWARD-VIEW UAV ROUTE VIEWER
前視無人機 3D 重建瀏覽器

CSV 載入飛行紀錄 (選填)
有 CSV 時使用 UTC 與 XYZ 校準; 沒有也能直接以照片建模。

IMG 載入前視影像
支援 JPG/PNG; 沒有 CSV 時依匯入順序建立拍攝路徑。

PLAN 載入工程圖底圖 (選填)
拖入平面圖、配置圖或施工圖圖片, 會鋪在路徑檢視底下。

工程圖顯示 尚未載入

透明度

縮放

旋轉 0

上

左

右

下

重設對位

節點密度 每 10 筆

航跡很長時可降低節點數, 點選 3D 節點會跳到對應時間。

本機重建

COLMAP COLMAP 4.1.0.dev0 (Commit 5b76f53 on 2026-04-27 with CUDA)

OpenMVS / Open3D 可執行

NVIDIA NVIDIA GeForce RTX 4070 Laptop

時間 6:11

資料筆數 4,883

距離 70.1 m

最高 Z 1.50 m

最大速度 0.84 m/s

照片 266

3D 2D Orbit Follow Onboard Fit

X 0.386 / Y 0.421 / Z -0.041

原始照片 點雲 貼圖模型 混合 Gaussian

266 張 匯入模型 清除 匯出模型

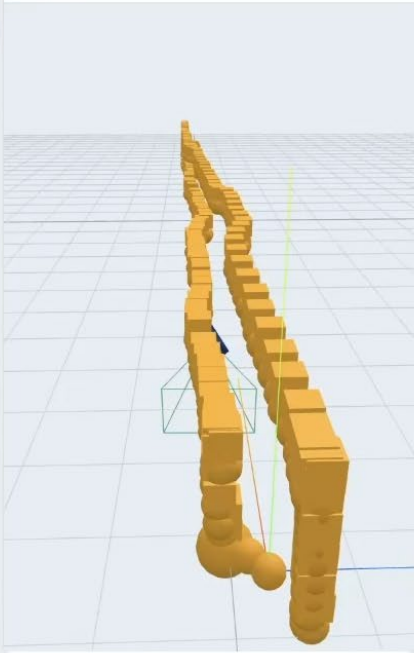


貼圖模型 / 403,817 triangles / 202,094 vertices / edge P99.9 3.27x

點選路徑查看畫面
在 3D 節點或 2D 工程圖航線上點一下, 前視畫面會直接跳到該位置。

目前時間附近資料

路徑定位



0:28 / 6:11
2026-07-29 02:30:25.952Z

365 / 4883

Model vs. Point Cloud Progress Analysis

YANKEY BIM / PLY 工程進度品質比對平台

File Edit View Window Help

YKE

YANKEY ENGINEERING

YANKEY ENGINEERING

BIM / PLY 工程進度品質比對平台 v2.0

● 等待幾何分析 產出巡檢報告

COORDINATE REGISTRATION

BIM / 無人機點雲自動座標對位

☐ 鎖定垂直軸 (僅水平旋轉) ☐ 控制點尺度校正 ☒ 粗略參考模式 自動對位

BIM 原始單位 點雲原始單位 點雲涵蓋範圍 點雲估計精度

英尺 (ft, Revit 常用) 公尺 (m) 模型最上方 1/3 ±200 mm

BIM 英尺 × 0.3048 ; 點雲 公尺 × 1 ; 統一換算為公尺 ; 估計精度 ±200 mm

1 匯入資料 2 自動對位 3 檢查並確認 4 執行偏差分析

信心 低信心	點雲近鄰率 (Fitness) 0.026 / 參考 0.368	Inlier RMSE 55.0 mm / 正式 11.7 ; 參考 300 mm	Holdout P50 / P95 55.2 / 70.8 mm	平移 XYZ 35.295 m / 16.166 m / 22.253 m	旋轉 XYZ / 尺度 -0.13° / -3.25° / -157.04° ; ×1.000000
-----------	-------------------------------------	--	-------------------------------------	--	--

原始座標 對位後 框選 BIM 區域 框選點雲區域 加入 BIM 控制點 加入點雲控制點 三點補修 清除

BIM ROI -- / 點雲 ROI -- ; BIM 0/3 , 點雲 0/3 確認對位

對位失敗, 已禁止偏差分析。請縮小共同區域或使用三組對應控制點。單位: BIM 英尺 × 0.3048、點雲 公尺 × 1 ; 配準範圍: 模型最上方 1/3。正規化尺寸: BIM 41.25 × 24.28 × 6.60 m ; 點雲 47.99 × 37.31 × 5.62 m。

SCAN TO BIM COMPARISON

模型與點雲疊合檢視

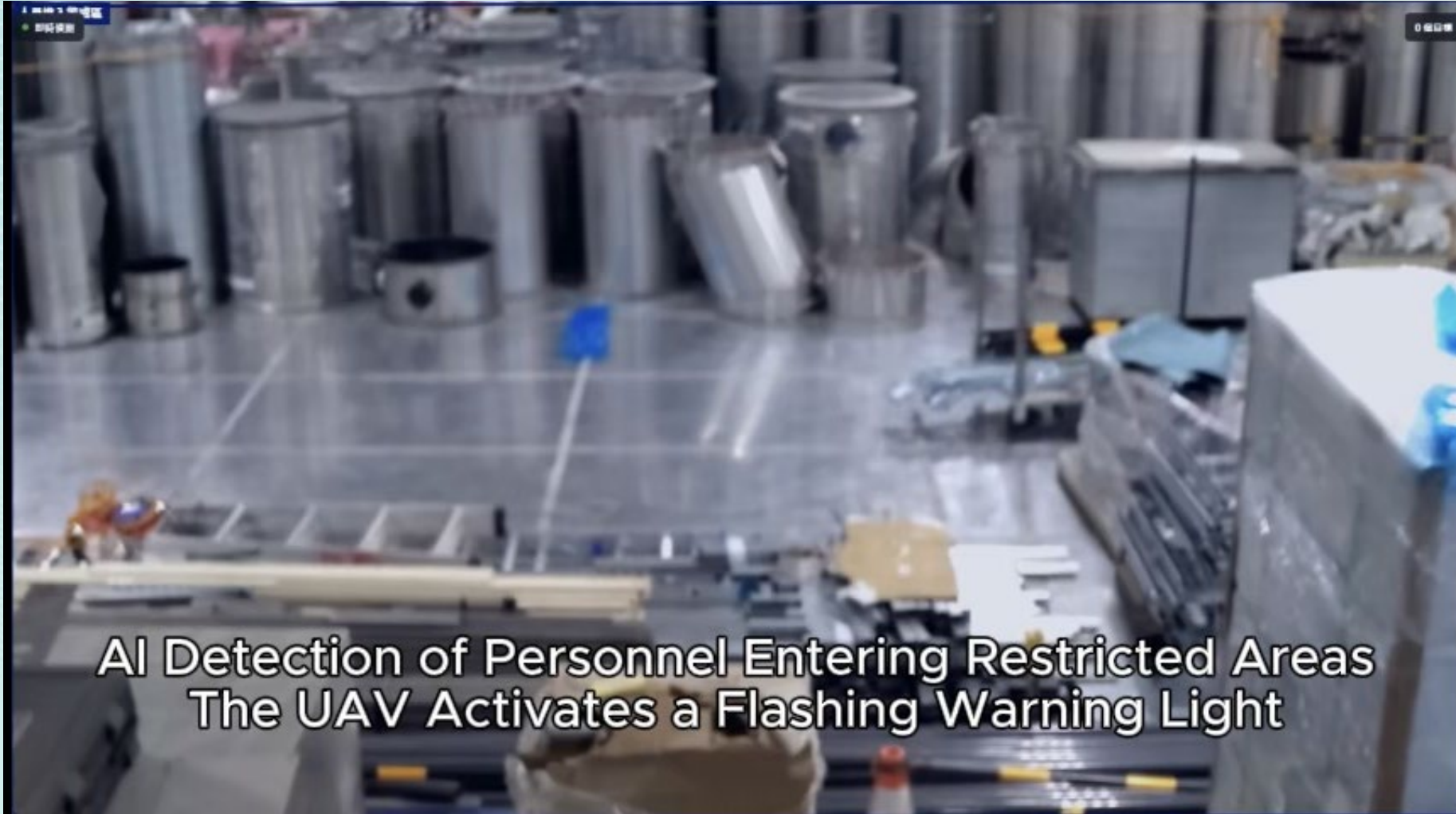
Model Point Cloud Overlay Deviation

AI ANOMALY DETECTION

即時偏差與風險 0 項

尚無真實比對結果。匯入 FBX 與 PLY 後執行區域比對。

EHS Safety Patrols: PPE Inspections and Restricted Zone Alerts





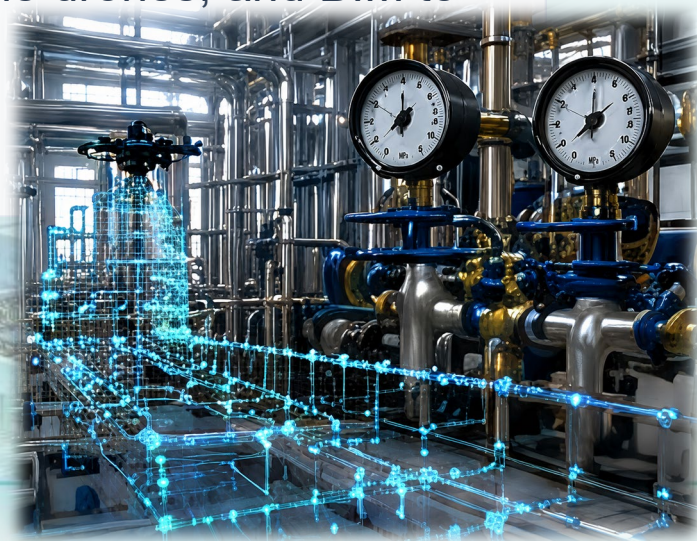
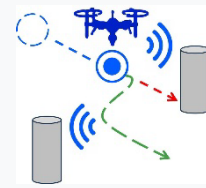
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AI UAVs in Action: Indoor Flight

Key Takeaways

Building a Smarter, Safer Construction Site

- **From "Auxiliary" to "Core" Data Foundation:** Elevating UAVs from simple photography to a critical 3D spatial data foundation
- **Dual-Track Integration Outdoor & Indoor Coverage:** Seamlessly combining outdoor 3D modeling with indoor AI inspections
- **Proprietary Made-in-Taiwan Tech Overcomes Key Pain Points:** Partnered with Avilon; leveraging vSLAM to eliminate indoor GPS blind spots
- **Multi-Dimensional Smart Safety Network:** Unifying fixed AI cameras, mobile drones, and BIM to uphold Yankey Engineering's high standards



Future R&D



Model Y



Prototype 2026



Model K



Model E

Future Outlook

Digital Twin Construction Site

Real-Time Synchronization: Integrates high-precision 3D models with IoT sensors for continuous live-site monitoring

Automated AI Progress & Deviation Tracking

Cloud-Driven Inspection: Cloud computing automatically compares point clouds against BIM to detect schedule delays and quality defects

Routine Autonomous Docking Operations

Unmanned Fleet Management: Leverages the Avilon Nest automated dock for fully autonomous, hands-free safety and progress patrols

ESG-Driven Sustainable Construction

Carbon Reduction & Compliance: Reduces carbon footprint and commuting risks, directly supporting YKE ESG and compliance targets



Science Technology





洋基工程股份有限公司
YANKEY Engineering Co., Ltd.

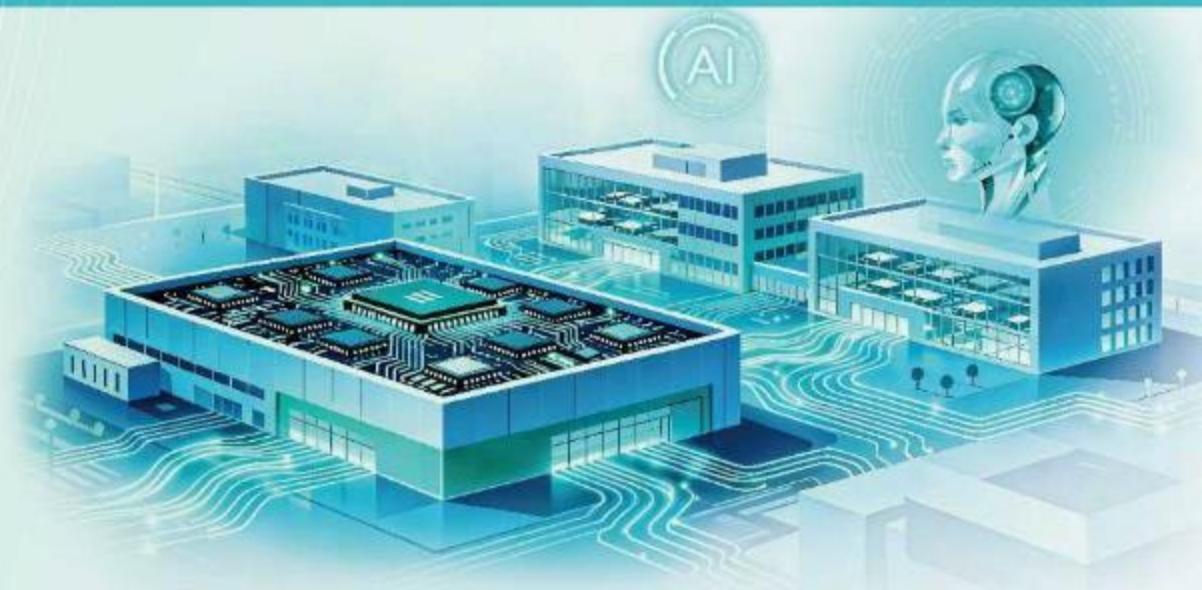
EPC Turnkey Engineering Services: CSA, MEP, and Cleanroom Systems
High-Tech Facilities, Industrial Plants, and Commercial Buildings





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Supporting Information



Key Challenge: Indoor Modeling & Inspection Gap

Challenge

Up to 85% of critical construction takes place indoors — making it the single largest blind spot for smart safety systems

Piping installations, structural interiors, and floor-level works demand high-frequency, full-coverage tracking, yet conventional drones are completely grounded w/o GPS

The Indoor Defense Line to Be Filled

■ The Main Battleground Is Indoors

Most critical, high-risk construction activities occur inside enclosed spaces.

■ GPS-Denied Indoor Limitation

Standard outdoor UAVs cannot navigate or hold position without satellite signals.

■ Demanding Spatial Challenges

Narrow corridors, tight piping, and physical obstacles limit visibility and mobility.

Yankey Engineering Solution:
Deploying indoor-positioning drones (non-GPS dependent) to fully bridge the indoor inspection and 3D modeling gap.



vSLAM Indoor Autonomous Flight: Key Specs & Adoption Benefits

01 Hardware & Compute

15 TOPS

Vision camera, IMU, compute module-Reference
5.32 TOPS ~ 21 TOPS

02 Response & Obstacle Avoidance

Brakes in 5–10 s

Safe avoidance distance: 1.8 m
Braking triggered after obstacle detection

03 Positioning Accuracy

1 ~ 10 cm

vSLAM visual position w/ QR Code
≈ 5 m place 1 positioning marker

04 Data Processing

**Real-time AI /
Offline Modeling**

AI recognition in real time on-device
Modeling done on a computer after data upload

Cost-Benefit: 3-Year Investment Comparison

Outsourced flight  **2.3M**

Equipment adoption  **1.4M**

Approximately 900K in savings; route configuration and equipment maintenance are required

Hurdles for Large-Scale Practical Use

- Positioning aid
- On-site safety
- Space constraints

Avoidance decisions rely on visual recognition of object color and texture similar to the background

Expected Benefits of Indoor AI-Driven UAV Deployment



Significant Reduction in Safety Risks

Eliminate Inspection Blind Spots:

Mobile AI detects real-time anomalies to proactively prevent workplace incidents



Marked Improvement in Inspection Efficiency

Automate Workflows:

Autonomous flight reduces patrol times while seamlessly integrating multi-sensor data



Full Visibility in Project Management

Visualize & Track Progress:

Overlays 3D point clouds with BIM models for objective, data-driven progress reviews

